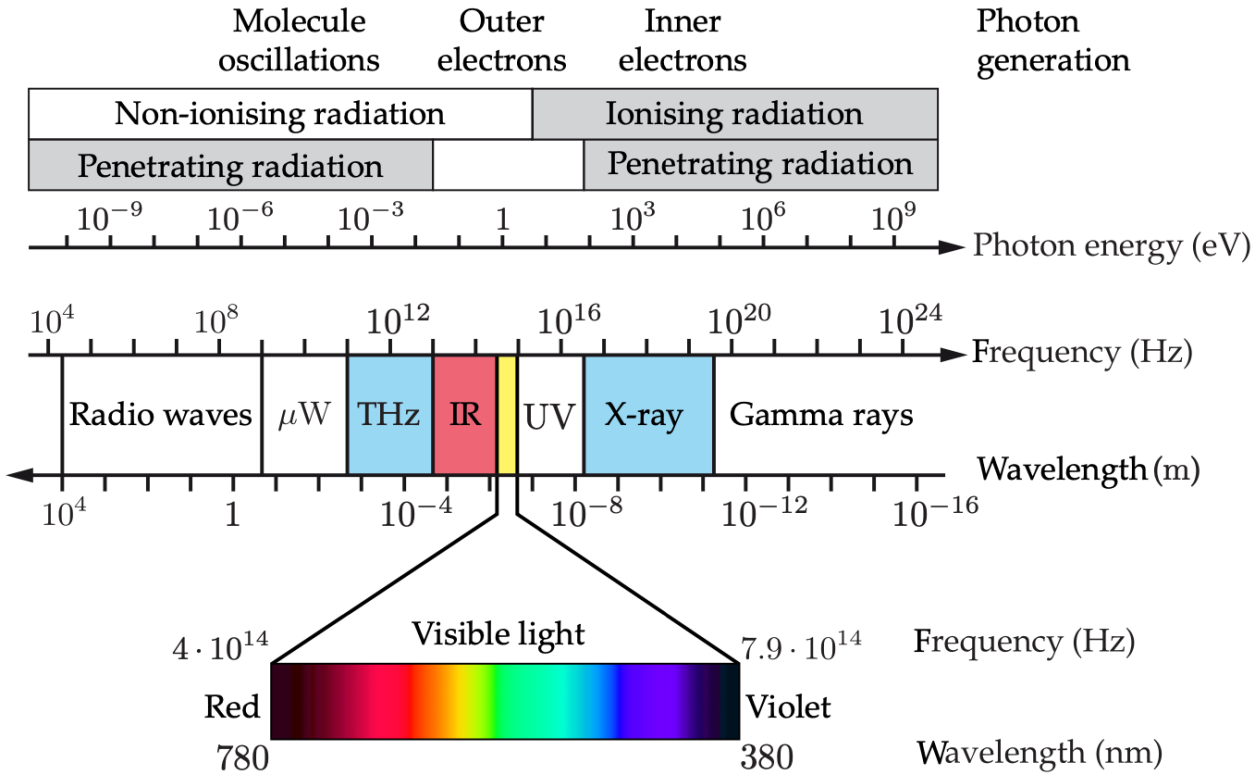
Computer Vision

Color

Technische Hochschule Rosenheim
Winter 2026/27
Prof. Dr. Jochen Schmidt

Electromagnetic Spectrum

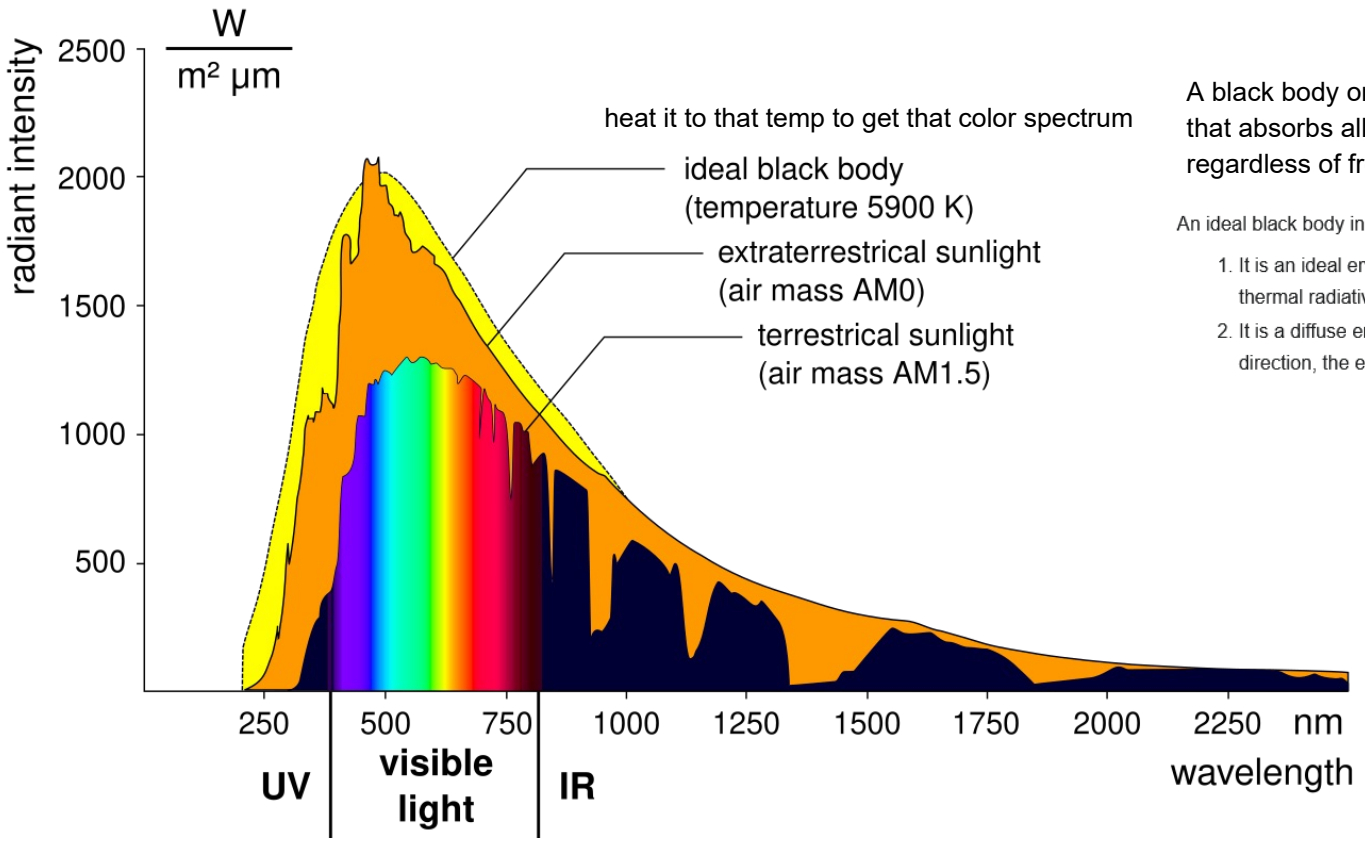


$\lambda = c/f$ f = frequency [Hz] λ = wavelength [m]
c = speed of light; in a vacuum, $c \approx 2.9979 \cdot 10^{8}$ m/s

μW = microwaves, THz = Terahertz radiation, IR = infrared, UV = ultraviolet

from: [Beyerer16]

Solar Spectrum



A black body or blackbody is an idealized physical body that absorbs all incident electromagnetic radiation, regardless of frequency or angle of incidence

An ideal black body in thermal equilibrium has two main properties:[2]

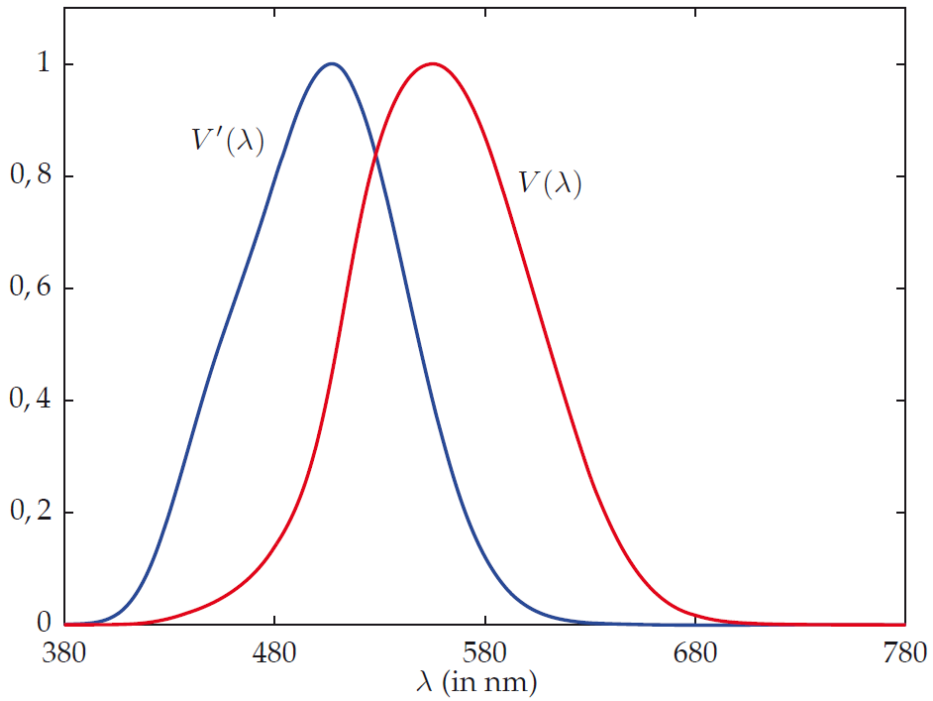
1. It is an ideal emitter: at every frequency, it emits as much or more thermal radiative energy as any other body at the same temperature.
2. It is a diffuse emitter: measured per unit area perpendicular to the direction, the energy is radiated isotropically, independent of direction.

The original uploader was Degreen at German Wikipedia. Improved Baba66 (opt Perhelion) on request; En. translation Locusta Fr. translation Eric Bajart Nl. translation BoH, Sonne Strahlungsintensitaet, CC BY-SA 2.0 DE

Perception of Brightness by the Human Eye

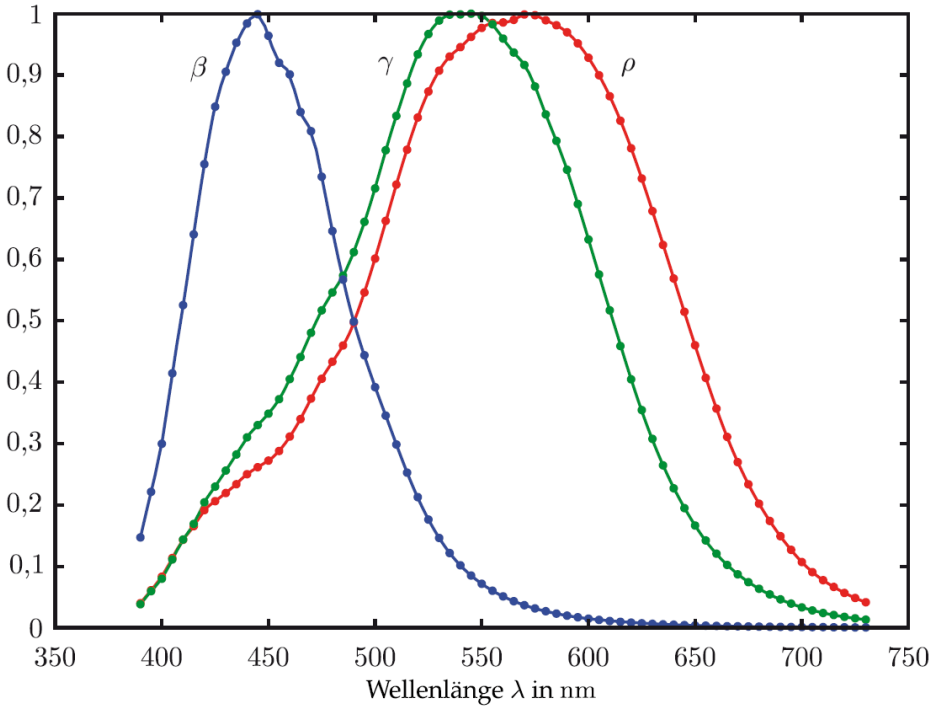
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Sensitivity of the human eye to brightness



V': dark-adapted eye
V: light-adapted eye

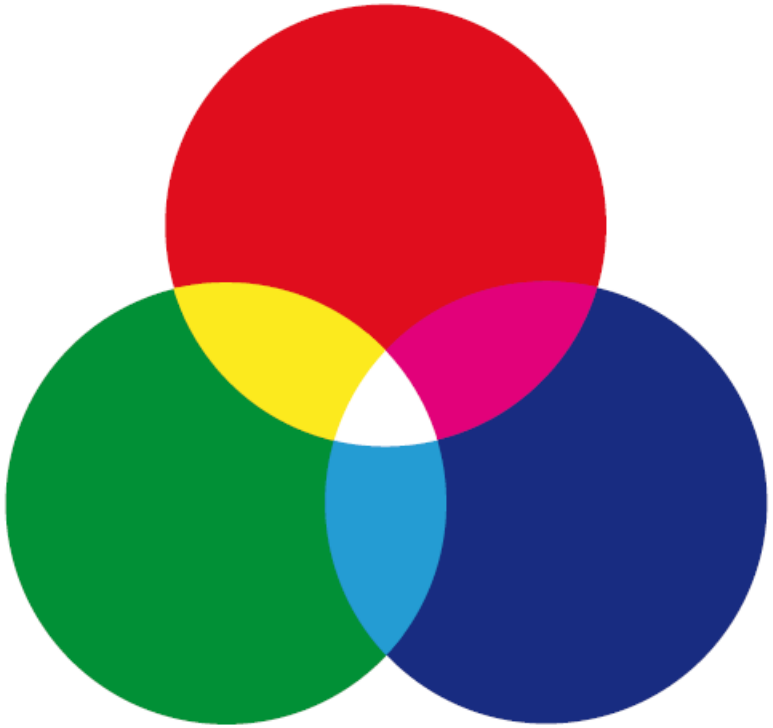
Spectral sensitivity of the cone cells



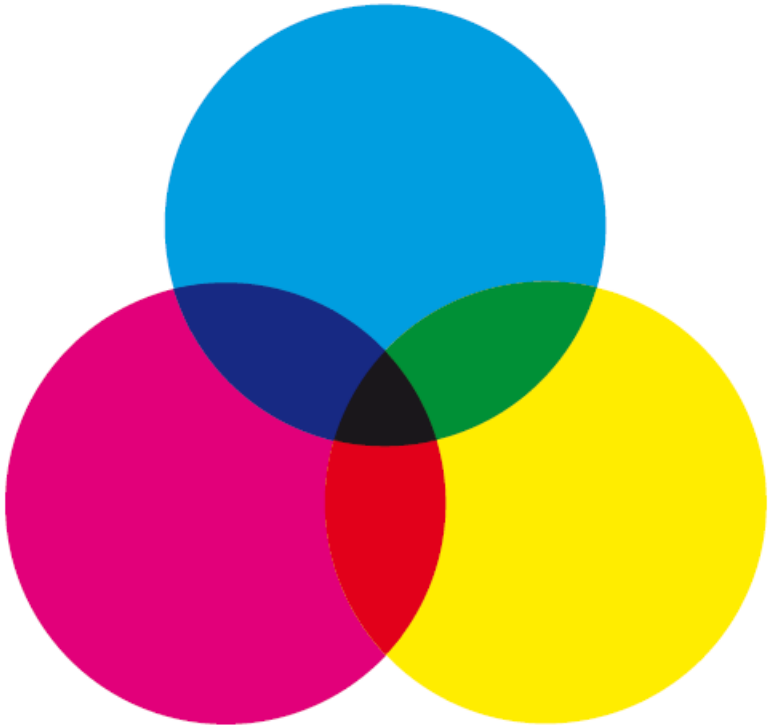
each curve normalized to maximum 1

from: [Beyerer16]

Additive and Subtractive Color Mixing



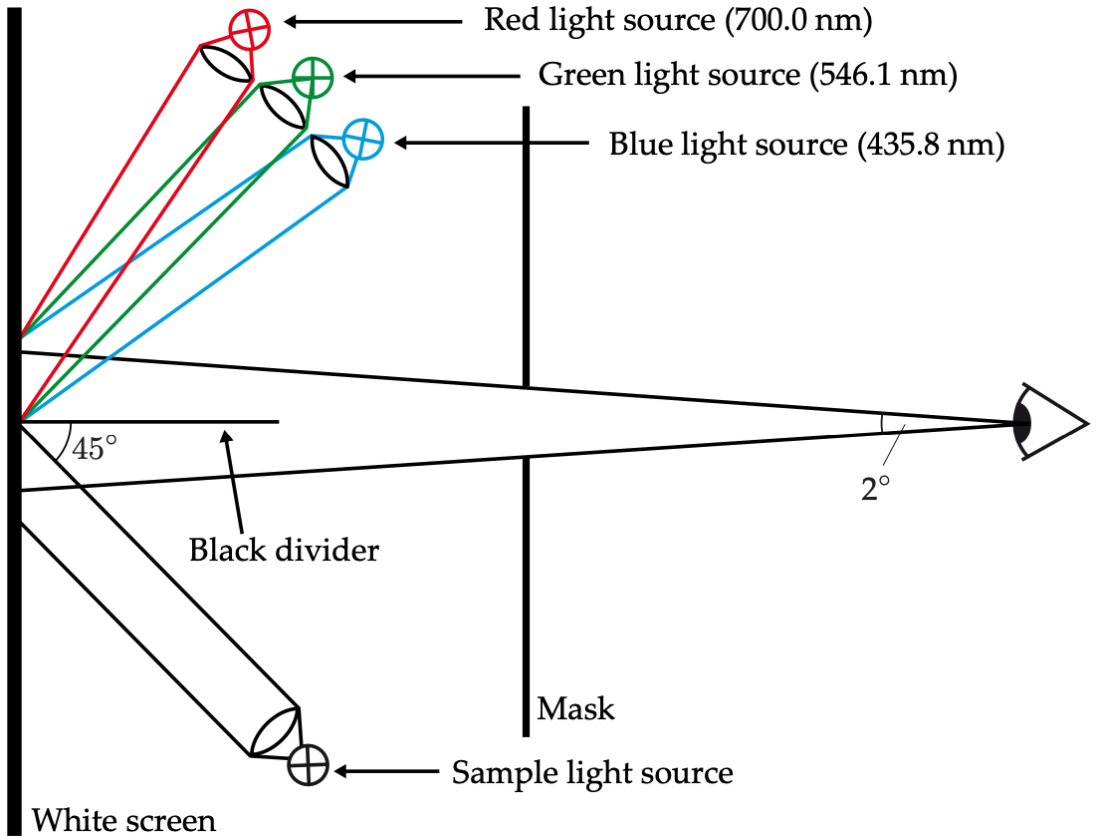
additive
real life



subtractive
used in printing

from: [Beyerer16]

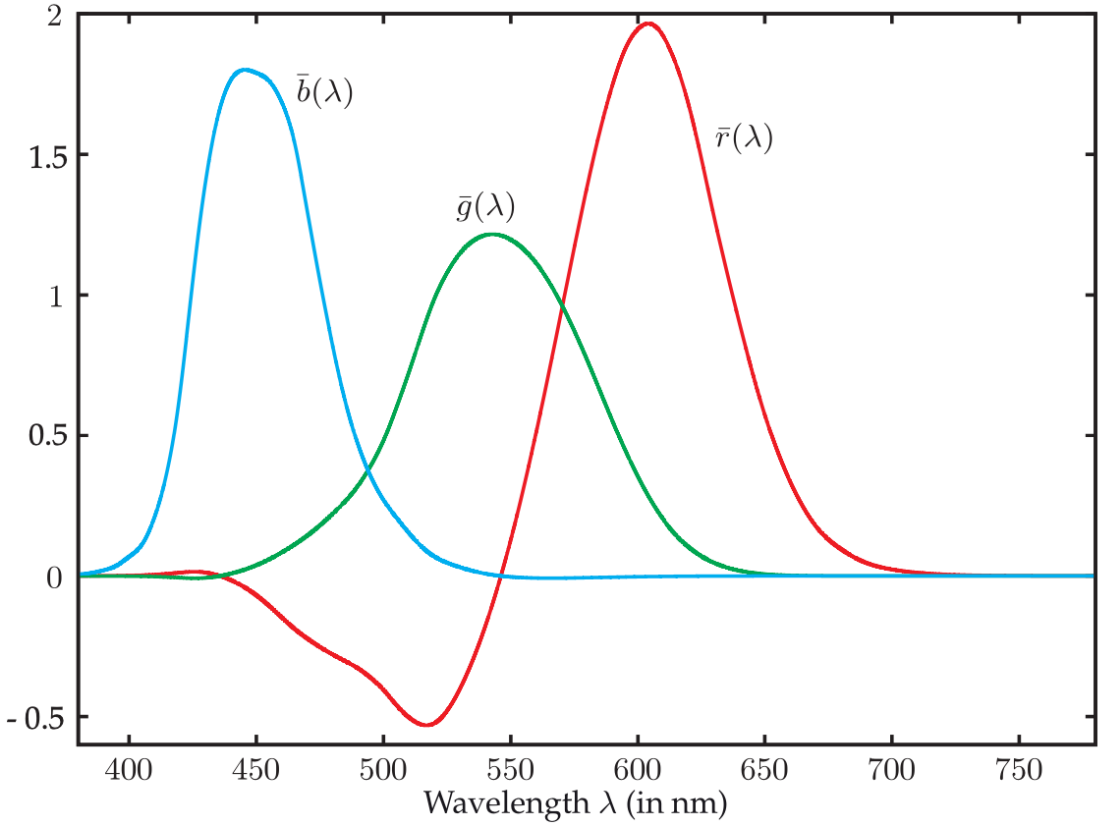
CIE Color Experiment



Color measurement experiment to empirically determine the color values with regard to the CIE primary colors

from: [Beyerer16]

Color Values of the Spectral Colors According to CIE



Scaled so that white light leads to

$R_{\mathrm{CIE}} = G_{\mathrm{CIE}} = B_{\mathrm{CIE}}$

Color value of any spectrum :

$$R_{\mathrm{CIE}} = \int_{380\mathrm{nm}}^{780\mathrm{nm}} I(\lambda)\bar{r}(\lambda)\mathrm{d}\lambda$$

$$G_{\mathrm{CIE}} = \int_{380\mathrm{nm}}^{780\mathrm{nm}} I(\lambda)\bar{g}(\lambda)\mathrm{d}\lambda$$

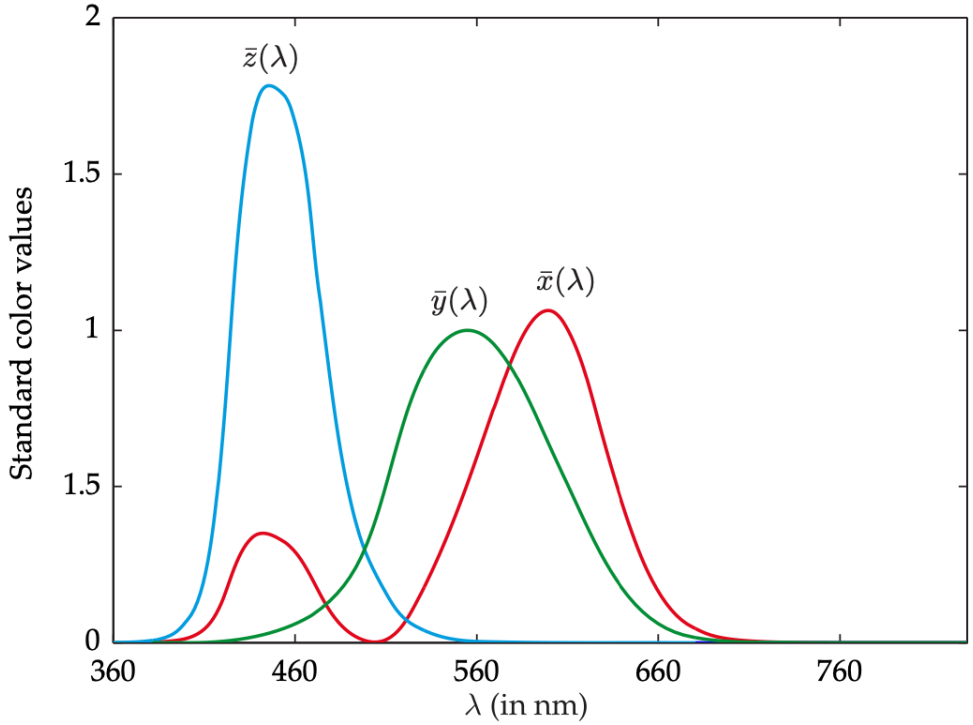
$$B_{\mathrm{CIE}} = \int_{380\mathrm{nm}}^{780\mathrm{nm}} I(\lambda)\bar{b}(\lambda)\mathrm{d}\lambda$$

there are certain colors that cannot be mixed
thus red (and green) goes into the negative which would theoretically allow the mixture of given colors

from: [Beyerer12]

CIE Standard Color Value Functions

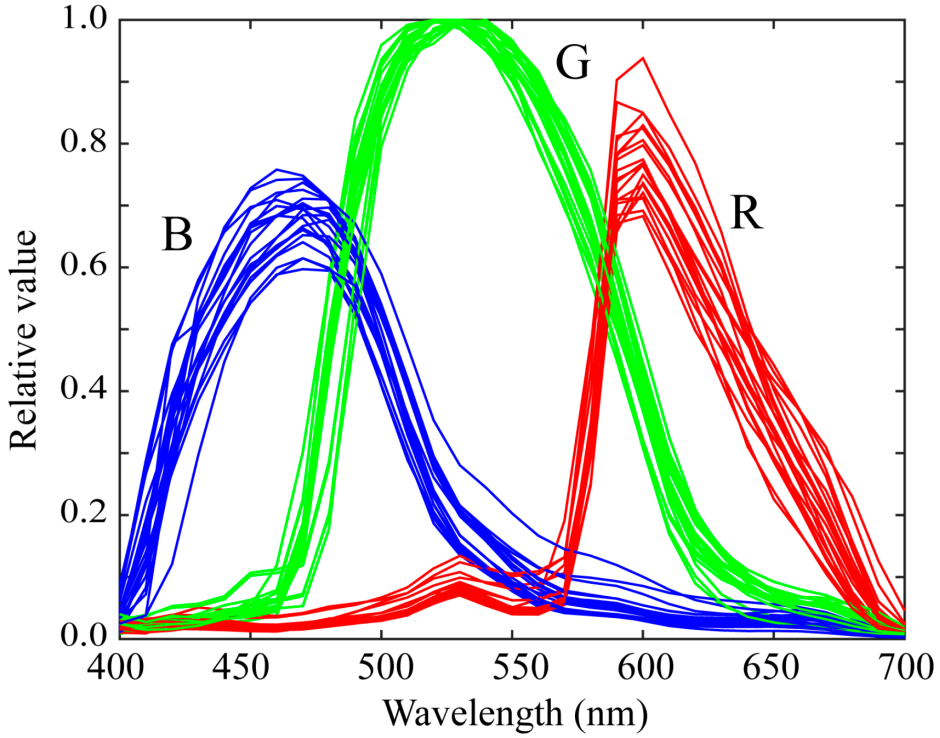
CIE standard color space
Color coordinates: X, Y, Z



transformed curves

$$\begin{pmatrix} X \\ Y \\ Z \end{pmatrix} = \begin{pmatrix} 0{,}49 & 0{,}31 & 0{,}20 \\ 0{,}17697 & 0{,}81240 & 0{,}01063 \\ 0{,}00 & 0{,}01 & 0{,}99 \end{pmatrix} \begin{pmatrix} R_{\text{CIE}} \\ G_{\text{CIE}} \\ B_{\text{CIE}} \end{pmatrix}$$

Y: Brightness = gray value image



spectral sensitivity of 20 mobile phone cameras

from: Tominaga, S.; Nishi, S.; Ohtera, R. Measurement and Estimation of Spectral Sensitivity Functions for Mobile Phone Cameras. *Sensors* **2021**, *21*, 4985. https://doi.org/10.3390/s21154985

CIE (x, y, z) Color Space

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Characterization of a color independent of brightness: 2D is sufficient → Normalization

Calculation of the normalized color value components x, y, z from standard color values X, Y, Z:

$$x = \frac{X}{X+Y+Z} \qquad y = \frac{Y}{X+Y+Z} \qquad z = \frac{Z}{X+Y+Z}$$

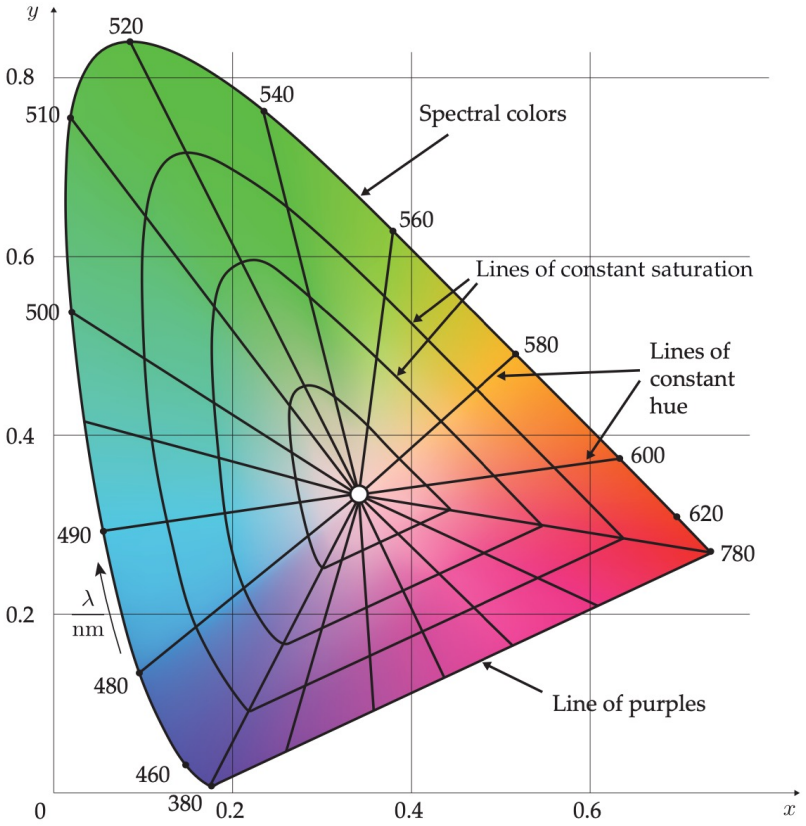
As $x + y + z = 1$: one color component is redundant → z discarded

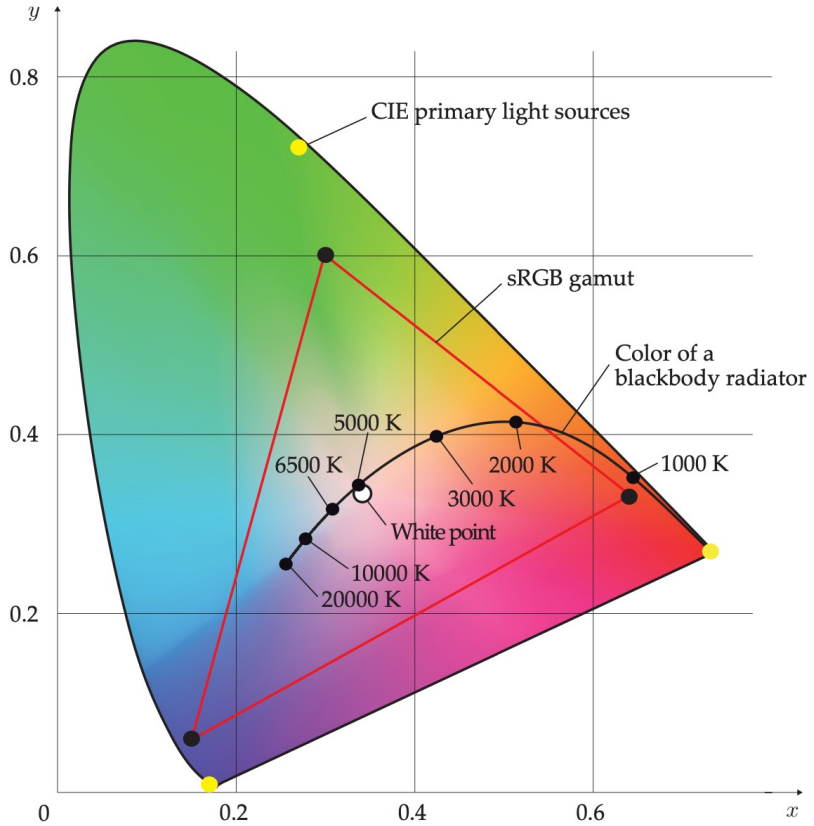
x, y: Chromaticity coordinates: describe color independent of brightness
Y: Luminance coordinate

(x, y, Y) completely describes a color value

CIE color space is a set of mathematical models to map human color perception to physical wavelengths of light

CIE Standard Color Chart (Chromaticity Diagram)





Additive color mixing:

- given:
  Chromaticity coordinates of N colors
- by additive mixing we get all colors within the convex hull (red line)

x/y is color
third dimension z (not shown here) would be brightness (which is why black is not shown here)

from: [Beyerer16]

CIE Standard Color Chart – Restrictions

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- Standard color values x, y cannot be interpreted easily (e.g. as hue or saturation)

- Distance between two points in the color chart ≠ Distance (similarity) of the associated colors
  - resolution of green is very fine
  - resolution of blue is very coarse
  - a certain distance in the x-direction can appear visually significantly smaller than the same distance in the y-direction

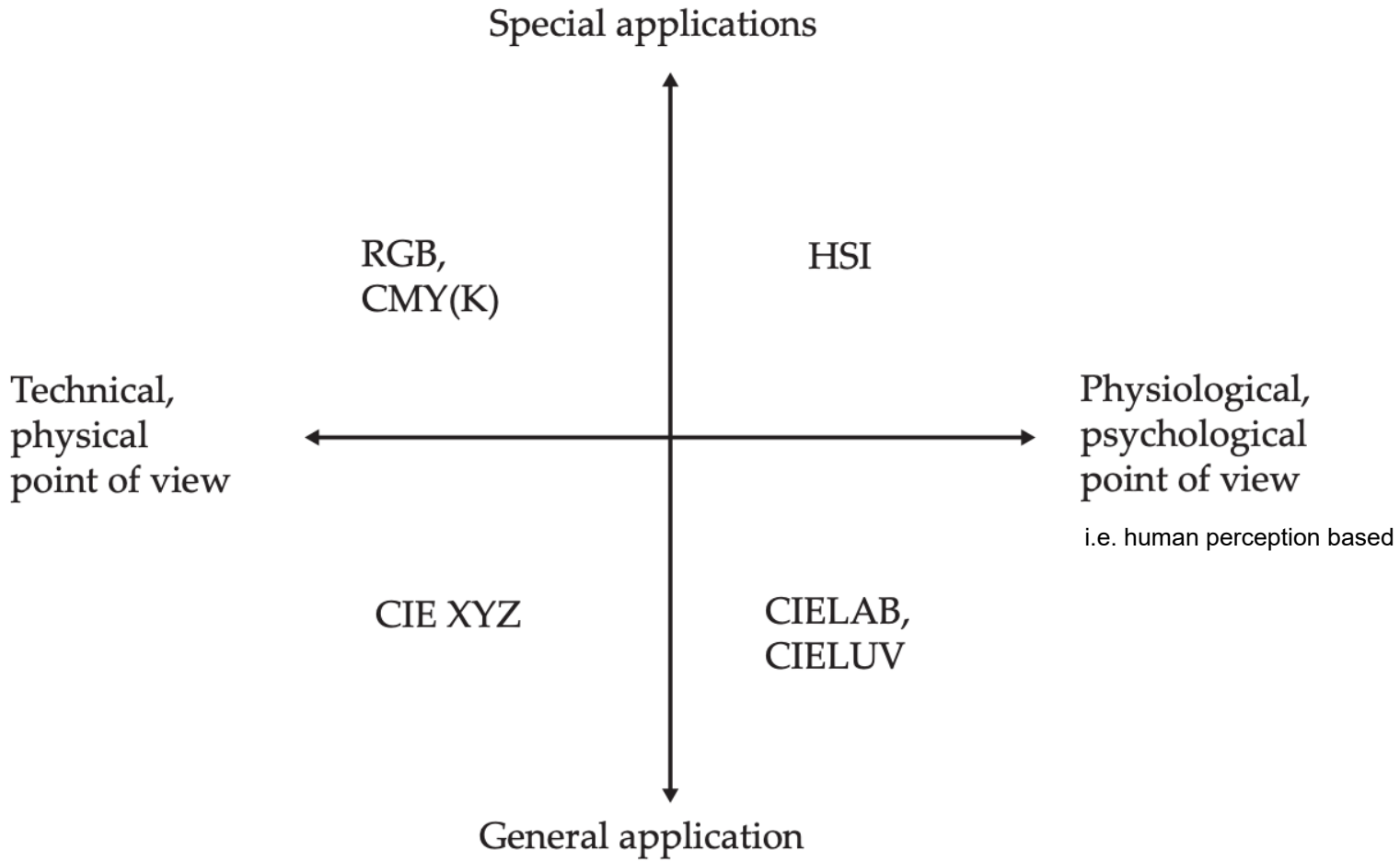
- Position of perfect black is not well-defined (because of $X = Y = Z = 0 \rightarrow$ denominator)

or rather not defined at all due to 0 division

$$x = \frac{X}{X + Y + Z}$$

→ Depending on the application, other color spaces are more suitable

Color Spaces – Classification



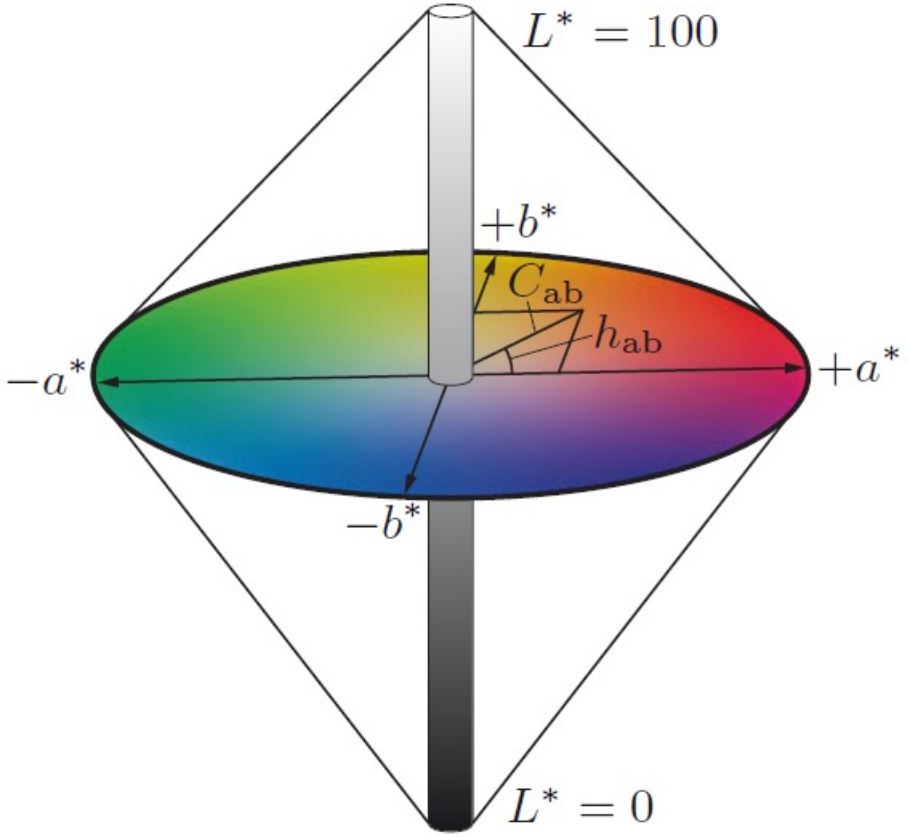
from: [Beyerer16]

CIELAB Color Space

- results from non-linear transformation
- L^*: Brightness
- a^*, b^*: Color differences to green component

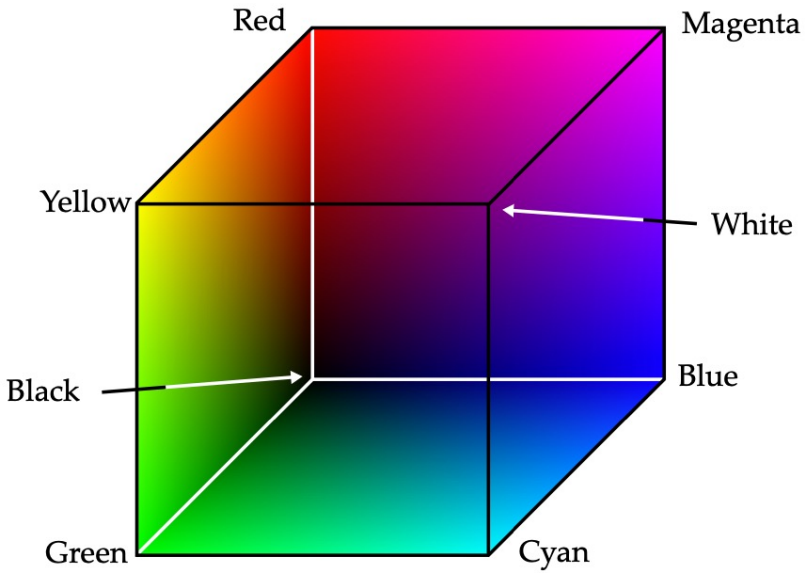
- colors arranged in a circle around the white point ($L^* = 100, a^* = b^* = 0$)
- achromatic colors: $a^* = b^* = 0$
- black: $L^* = 0, a^* = b^* = 0$

- Polar coordinates *C*, *h* illustrative:
  - h = hue angle
  - C = radius (chroma) = saturation



from: [Beyerer16]

RGB Color Space



- additive color mixing:
  Red (R), Green (G), Blue (B)
- Value range: 0 – 1

- Used for: Monitors, cameras, projectors
- RGB color spaces and color gamuts vary depending on the device
- device-independent: sRGB (**s**tandard **RGB**)

from: [Beyerer16]

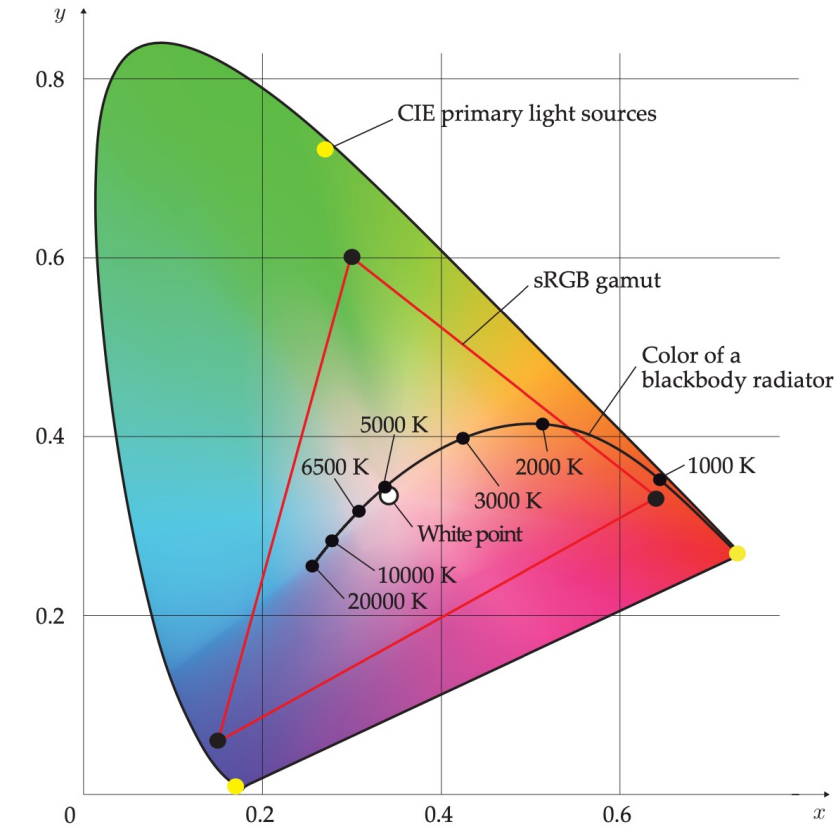
sRGB – linear

is obtained from XYZ space by

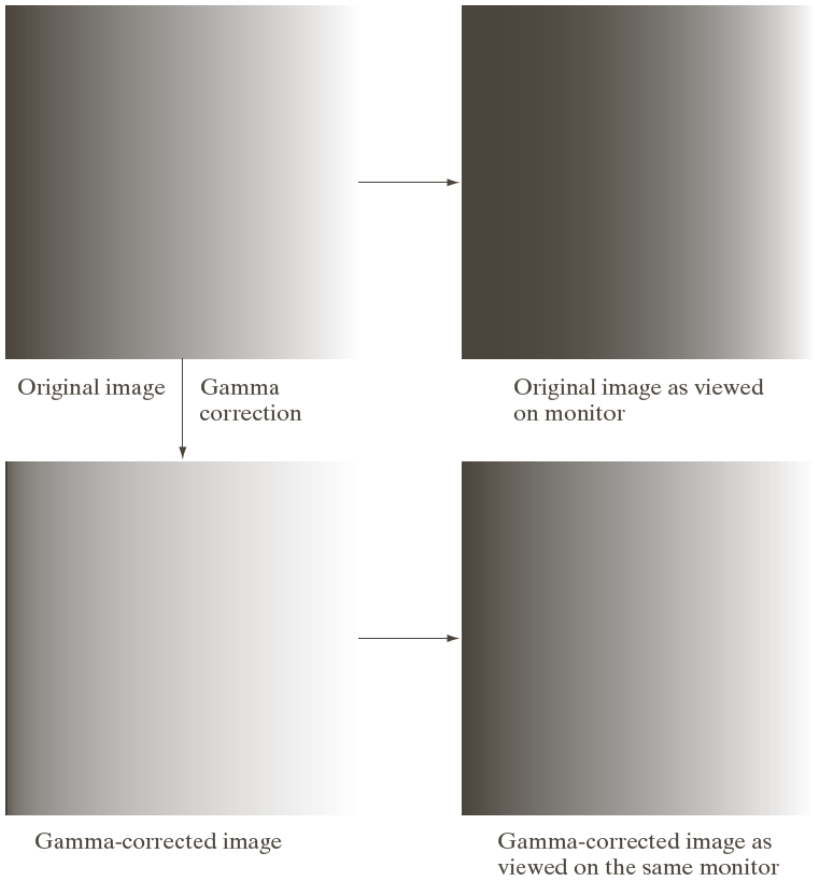
$$\begin{pmatrix} R_s \\ G_s \\ B_s \end{pmatrix} = \begin{pmatrix} 3{,}2410 & -1{,}5374 & -0{,}4986 \\ -0{,}9692 & 1{,}8760 & 0{,}0416 \\ 0{,}0556 & -0{,}2040 & 1{,}0570 \end{pmatrix} \begin{pmatrix} X \\ Y \\ Z \end{pmatrix}$$

- if a value becomes negative → value = 0
- if a value > 1 → value = 1
- these colors cannot be represented in sRGB
- transformation sRGB to XYZ: invert matrix

- sRGB non-linear → results from linear sRGB by gamma correction



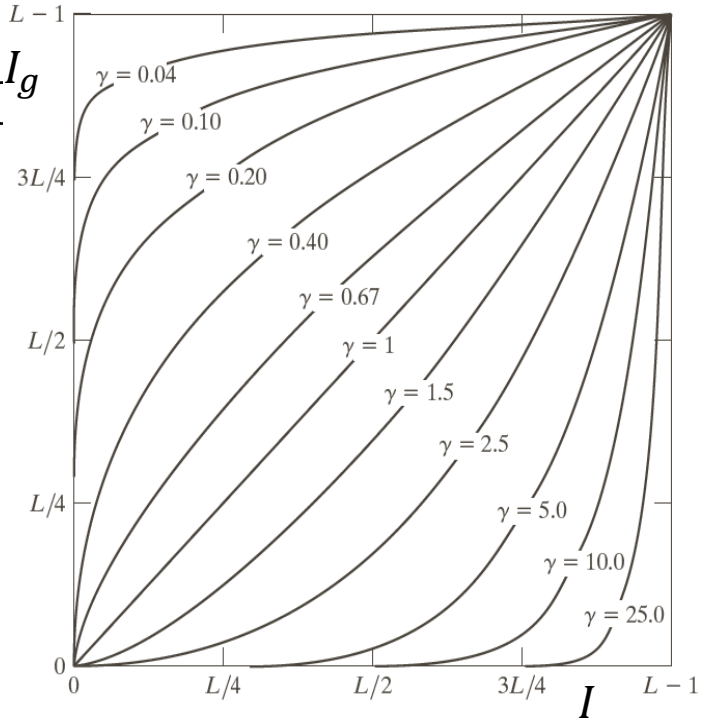
Gamma Correction



- Equalization of
  - Non-linearities in human perception
  - non-linear characteristics of devices (monitor, camera)
- Gamma correction of an intensity I:

$$I_g = cI^\gamma$$
$$c, \gamma > 0, \qquad I \in [0$$

- often: $c = 1$
- Transformation is reversible



from: [Gonzalez08]

Gamma Correction

Original



$\gamma = 3$



$\gamma = 4$



$\gamma = 5$



$c = 1$

from: [Gonzalez08]

sRGB – non-linear

- usually: additional non-linear gamma correction on top of (linear) sRGB
- adjusts intensities to human perception
- improves display on monitors
- corrected sRGB values are still in [0; 1]

$$\begin{pmatrix} R_s' \\ G_s' \\ B_s' \end{pmatrix} = \begin{pmatrix} f_\gamma(R_s) \\ f_\gamma(G_s) \\ f_\gamma(B_s) \end{pmatrix} \qquad \text{with } f_\gamma(x) = \begin{cases} 1{,}055x^{\frac{1}{2{,}4}} - 0{,}055 & \text{if } x > 0{,}00304 \\ 12{,}92x & \text{if } x \leq 0{,}00304 \end{cases}$$

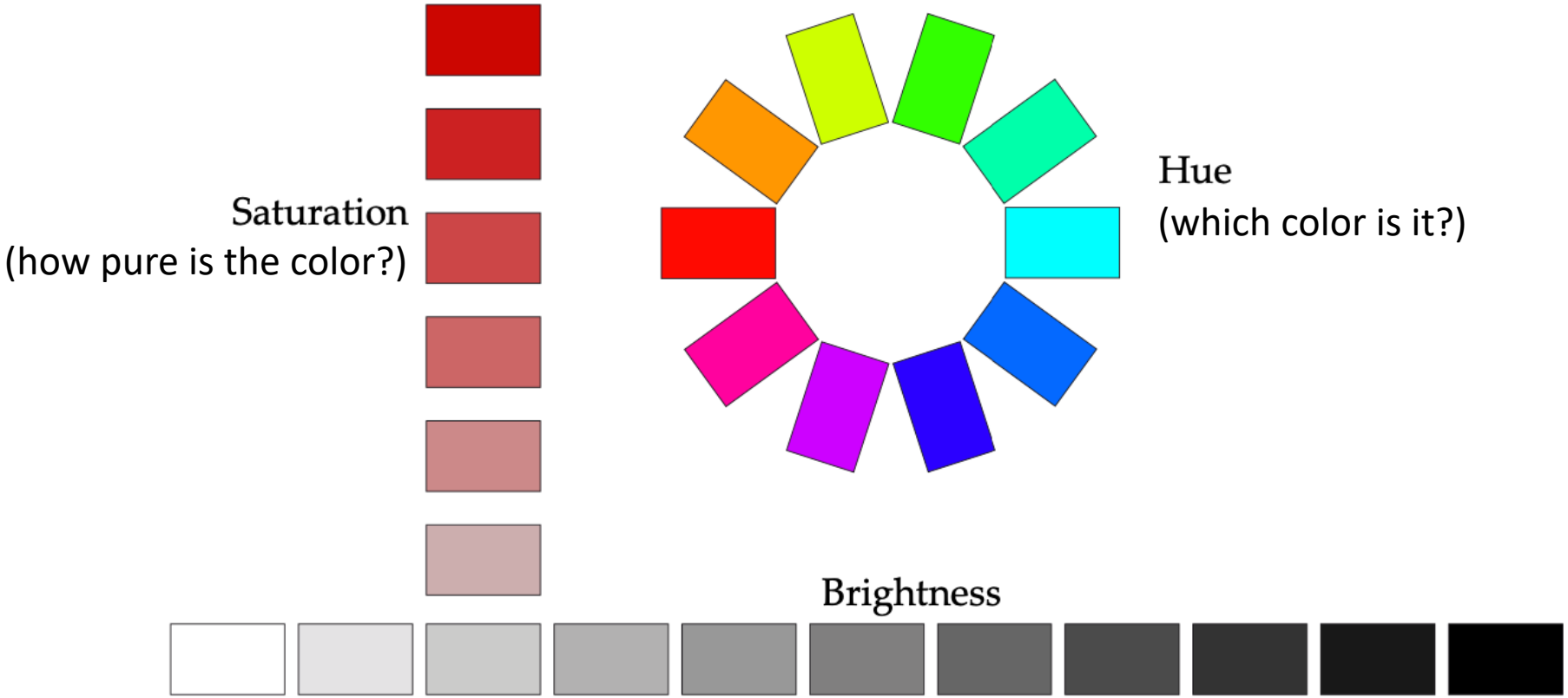
HSV/HSL/HSI

- H = Hue (*Farbton*)
  S = Saturation (*Sättigung*)
  V = Value, L = Luminance, I = Intensity (Brightness, *Helligkeit*)
  - all three color spaces are slightly different
  - values can be easily interpreted
- Intensity coordinate
  - weighs all channels equally
  - does not correspond to the human perception of brightness

- Calculation of HSI from gamma-corrected sRGB:

$$H = \arctan \frac{\sqrt{3}(G'_s - B'_s)}{2R'_s - G'_s - B'_s} \qquad I = \frac{1}{3}(R'_s + G'_s + B'_s) \qquad S = 1 - \frac{\min\{R'_s, G'_s, B'_s\}}{I}$$

Hue, Saturation, Brightness



from: [Beyerer16]

YUV/YC_BC_R

- Brightness (Y) and color difference signals (U, V / C_B, C_R)
- Usage
  - YUV: PAL television standard
  - YC_BC_R: JPEG

- Calculation from gamma-corrected sRGB:

$$Y = 0{,}299R'_s + 0{,}587G'_s + 0{,}114B'_s$$

$$U = 0{,}493(B'_s - Y) \qquad C_B = 0{,}564(B'_s - Y)$$

$$V = 0{,}877(R'_s - Y) \qquad C_R = 0{,}713(R'_s - Y)$$

better compressable, closer to human perception which is more perceptive to brightness changes than color changes, meaning preserving brightness is more important than color
this in turn means that it is possible to remove lots of color information (compression) without being noticeable to the human eye
color space subsampling 4:1:1 > e.g. 4:4:4 keep every YUV value, 4:2:2 skip every second one

Sources

[Beyerer16] Beyerer, J., Puente Leon, F., Frese, Ch.: ***Machine Vision***, Springer, 2016.

[Gonzalez08] Gonzalez, R.C., Woods, R.E.: ***Digital Image Processing***, Prentice Hall International, 3rd edition, 2008.

conclusion: depending on the application and use case, different color spaces are needed